

LUCKY SAMUEL

Backend Engineer | C# / .NET Developer | Full-Stack Capable

Email: lucky.samuel.egiagbo@gmail.com | Location: Lagos, Nigeria | LinkedIn: linkedin.com/in/lucky-samuel
GitHub: github.com/alchemistlowkey | Portfolio: lucky-samuel.vercel.app

PROFESSIONAL SUMMARY

Results-driven Backend Engineer with 3+ years of experience designing, developing, and maintaining production-ready web applications and distributed systems using C# and ASP.NET Core (.NET 5–10). Proven track record delivering medium-to-large-scale, high-volume, database-driven web applications end-to-end — from architecture through deployment. Deep expertise in Clean Architecture, Microservices, CQRS with MediatR, and RESTful API design. Solid command of Microsoft SQL Server, PostgreSQL, and MongoDB with Entity Framework Core and Dapper. Experienced integrating external web portals and third-party APIs. Frontend-capable in SvelteKit, React, and Next.js. Thrives in Agile/Scrum team environments; comfortable working independently to meet deadlines. Passionate about clean, testable, and maintainable code.

TECHNICAL SKILLS

Languages: C#, JavaScript, HTML5, CSS3

Frameworks & Platforms: ASP.NET Core, .NET 5–10 (Minimal APIs), Node.js, Express.js

Architecture & Patterns: Clean Architecture, Vertical Slice Architecture, Microservices, CQRS / MediatR, Repository Pattern, Dependency Injection, Event-Driven Architecture, SOLID Principles, OOP

Databases: Microsoft SQL Server (MSSQL), PostgreSQL, MongoDB — Entity Framework Core, Dapper

Messaging & Queues: RabbitMQ, MassTransit, Message Queues, Distributed Systems

Frontend: SvelteKit, React, Next.js, Tailwind CSS

Testing: xUnit, Moq, FluentAssertions, Unit Testing, Integration Testing (90%+ coverage achieved)

Auth & Security: JWT, Keycloak, Role-Based Authorization (RBAC), OAuth 2.0 / OIDC

Tools & DevOps: Docker, Docker Compose, Git, GitHub, Swagger / OpenAPI / Scalar, Serilog, AutoMapper, FluentValidation, YARP, Agile / Scrum

Cloud & Hosting: Vercel, Railway, AWS (basic), Azure (basic)

Payment Integrations: Paystack, Flutterwave (webhook handling, error handling)

Media & Storage: Cloudinary (per-tenant scoped uploads)

PROJECTS

HealthBridge — Multi-Tenant Healthcare Management Platform

.NET 10 · ASP.NET Core Minimal APIs · MongoDB · RabbitMQ / MassTransit · Keycloak · Docker · Serilog / Seq · YARP · Cloudinary | 2025 – Present

- Architecting and building a cloud-native, multi-tenant healthcare management platform for hospitals, patients, and HMO organisations in Nigeria/West Africa, targeting 50+ hospital tenants and sub-200ms P95 API response times.
- Designed a 7-service Microservices topology (Hospital, Patient, HMO, Appointment, Billing, Notification, Gateway) each internally structured with Vertical Slice Architecture and CQRS + MediatR — every feature is a self-contained slice with co-located handler, validator, request/response DTOs, and endpoint definition.
- Implemented a robust self-service tenant onboarding system: auto-generates unique tenant slugs via atomic MongoDB sequence increments (FindOneAndUpdate + \$inc), records ownership in a tenant_registry, and stamps tenant_id onto the user's Keycloak account via the Admin REST API — with full rollback if the Keycloak call fails, keeping the operation safely retryable.
- Resolved a critical multi-tenancy correctness bug: enforced a single authoritative JWT-resolved tenant source (ITenantContext) across all repository queries, eliminating silent cross-collection data leakage between write and read paths.
- Wired Keycloak JWT authentication with correct role claim mapping from nested realm_access.roles; configured Swagger with both Bearer and OAuth2 security schemes and EnablePersistAuthorization() for reliable API testing.
- Built event-driven Notification Worker consuming RabbitMQ domain events (appointment booked, invoice generated, patient enrolled, etc.) via MassTransit consumers — purely event-driven background service with dead letter queue configured.
- Integrated Cloudinary for per-tenant scoped media uploads (hospital logos, patient profile pictures) and Serilog + Seq for structured observability across all services.
- Milestones completed: Infrastructure, Hospital, Patient, HMO, Appointment, Billing, and Notification services. Gateway (YARP), integration tests, and full documentation pending.

E-Commerce Platform — Production-Ready Web Application

ASP.NET Core 10 (C#) · PostgreSQL · Entity Framework Core · SvelteKit · Paystack · Flutterwave | 2025 – Present

- Built a production-ready, high-volume e-commerce web application using ASP.NET Core 10 with Clean Architecture (Repository & Service patterns) and PostgreSQL with EF Core migrations.
- Implemented JWT authentication; integrated Paystack and Flutterwave payment gateways with robust error handling and webhook processing; added Serilog structured logging and Scalar OpenAPI documentation.

- Developed SvelteKit storefront for product catalogue, cart management, and order processing with real-time cart synchronisation.

Play Microservices — Event-Driven Distributed System

.NET 5 · ASP.NET Core · RabbitMQ · MassTransit · MongoDB · Next.js 15 · Docker | 2026

- Architected an event-driven microservices system using .NET 5 with RabbitMQ and MassTransit for asynchronous inter-service communication.
- Built Catalog Service (RESTful CRUD) and Inventory Service (event consumer); created shared Play.Common library with reusable MongoDB repositories and MassTransit extensions; containerised with Docker Compose.

BookStore API — Vertical Slice Architecture with CQRS

ASP.NET Core · .NET 10 · MediatR · Keycloak · xUnit · Moq · FluentValidation | 2024

- Developed a scalable RESTful Web API using Vertical Slice Architecture with MediatR for CQRS — each feature is a self-contained slice with co-located handler, validator, and DTOs.
- Implemented JWT authentication via Keycloak with role-based authorization; achieved 90%+ test coverage with comprehensive xUnit + Moq suite.

Expense Tracker API — Full-Stack Web Application

ASP.NET Core 10 (C#) · PostgreSQL · Entity Framework Core · SvelteKit | 2024

- Built a Clean Architecture Web API for expense tracking with layered Repository/Service design, JWT auth with token refresh, and EF Core PostgreSQL persistence.
- Created Swagger/OpenAPI documentation; built SvelteKit frontend with real-time expense categorisation, filtering, and spending visualisation charts.

Invoicely — Invoicing SaaS Web Application

SvelteKit · Node.js · Tailwind CSS v4 | 2024 – Present

- Built a full-featured invoicing SaaS with client management, real-time PDF generation, and mobile-first responsive dashboard.
- Implemented shared Svelte writable store for cross-component state synchronisation and card-based UI alternatives to data tables.

EDUCATION

Bachelor of Science — Industrial Chemistry

University of Benin · Nigeria

Certificate of Completion — Back-End Specialisation (12-Month Programme)

ALX Software Engineering (Powered by Holberton Inc) · June 2024

CERTIFICATIONS

Career Essentials in Software Development · Microsoft & LinkedIn Learning · Jun 2024

Build Your Generative AI Productivity Skills · Microsoft & LinkedIn Learning · Jul 2024

Introduction to Prompt Engineering for Generative AI · LinkedIn Learning · Jul 2024

Programming Foundations: Beyond the Fundamentals · LinkedIn Learning · Jun 2024

Introduction to Career Skills in Software Development · LinkedIn Learning · Jun 2024

English for IT 1 · Cisco Networking Academy / OpenEDG · Jun 2024

English for IT 2 · Cisco Networking Academy / OpenEDG · Jun 2024

Professional Soft Skills Learning Pathway · LinkedIn Learning · Aug 2024

Communication Foundations · LinkedIn Learning · Aug 2024

Effective Listening · LinkedIn Learning · Jul 2024

Teamwork Foundations · LinkedIn Learning · Aug 2024

Critical Thinking for Better Judgment & Decision-Making · LinkedIn Learning · Jul 2024

Developing Your Emotional Intelligence · LinkedIn Learning · Jul 2024

Building Resilience · LinkedIn Learning · Jul 2024

Embracing Unexpected Change · LinkedIn Learning · Jul 2024

Unconscious Bias (CPE Credit — NASBA) · LinkedIn Learning · Jul 2024

Writing in Plain Language · LinkedIn Learning · Jul 2024

ADDITIONAL INFORMATION

- Excellent written and verbal communication skills; able to clearly articulate technical concepts to both technical and non-technical stakeholders.
- Experienced in Agile / Scrum team environments — sprint planning, code reviews, and iterative delivery.
- Self-motivated, detail-oriented, and organised; able to work independently and efficiently to meet deadlines.
- Collaborative team player with a willingness to teach and to learn within cross-functional engineering teams.
- Experience developing and integrating external web portals and APIs allowing users to input and retrieve accurate real-time information.